- An API gateway is a specialized reverse proxy that can make different APIs appear as if they were a single API. It is often used within a *microservices architecture* to simplify the API presented to clients. API gateways can often also take care of some of the aspects of API security discussed in this book.
- A web application firewall (WAF) inspects traffic at a higher level than a traditional firewall and can detect and block many common attacks against HTTP web services.
- An intrusion detection system (IDS) or intrusion prevention system (IPS) monitors traffic within your internal networks. When it detects suspicious patterns of activity it can either raise an alert or actively attempt to block the suspicious traffic.

> **DEFINITION** In a ***microservices architecture***, an application is deployed as a collection of loosely-coupled services rather than a single large application, or monolith. Each microservice exposes an API that other services talk to. Securing microservice APIs is covered in detail in part 3 of this book.

In practice, there is often some overlap between these services. For example, many load balancers are also capable of performing tasks of a reverse proxy, such as terminating TLS connections, while many reverse proxies can also function as an API gateway. Some more specialized services can even handle many of the security mechanisms that you will learn in this book, and it is becoming common to let a gateway or reverse proxy handle at least some of these